SoftWorld: Development and Preliminary Verification of a Disc-Based Soft-Limb Simulation with Virtual Strain Sensing

Research Development Record — Working Configuration

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Abstract

This report documents the development of a ROS 2/Isaac Sim platform for commanding a compliant disc-based limb, estimating bending strain from its measured orientations, and reconstructing its shape in RViz. The active configuration contains six discs, five disc-to-disc intervals, and two rings of three virtual strain gauges. Development included position and hybrid control scripts, hanging-base geometry, damping changes, independent pose-derived sensing, live strain plots, and migration from a second simulated limb to a sensor-driven visualization. A verified controller-topology mismatch caused a requested 180° bend to produce approximately 81.818°: commands retained the twelve-disc controller's eleven-interval normalization. The working loader now converts identified legacy position targets to the five-interval configuration. An end-to-end headless test using the original controller measured 179.9996° at the sixth disc. These results establish a bounded software regression result, not physical material validation or general constant-curvature accuracy. A prior development configuration verified a 270.0000° torque-admittance result using accumulated inter-disc bend measurement; the active working configuration is now intentionally limited to 180°. Large-angle diagonal orientation errors, sensor-model assumptions, and the absence of physical torque calibration remain explicitly identified limitations.

Document status. This is a research-oriented technical record, not a completed thesis, peer-reviewed publication, or institutionally approved experimental protocol. It distinguishes source-code facts, recorded numerical tests, user observations, historical requests, and proposed future experiments. Author identity, advisor, research objectives, and thesis-specific requirements must be supplied by the researcher. No physical experiment is claimed.

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1 Research context, objectives, and scope

The researcher identified this project as part of MSET research in ETID at Texas A&M University. The present engineering objective is to develop a repeatable simulation workflow for compliant limb actuation and orientation-derived shape sensing. The long-term concept is to reproduce the configuration of a leader limb from sensor readings without providing those leader commands to the reconstruction system.

The active workflow is:

Position or torque command
↓
Physical disc-chain leader in Isaac Sim
↓
Six orientation-derived virtual strain readings
↓
Local curvature reconstruction and interpolation
↓
`/joint_states` and robot-state publisher
↓
TF and sensor-driven limb visualization in RViz

The word "physical" above means the physics-simulated leader, not laboratory hardware. The RViz model is a kinematic visualization, not a dynamically actuated second robot. The earlier two-limb Isaac configuration is retained as a backup. Earlier quadruped work is recorded as development history; the quadruped is separate from the currently verified six-disc workflow.

2 Evidence and documentation provenance

This report is based on the development conversation, the repository README, inspection of the working Python files, and available verification logs. The supplied Webster–Jones review was consulted for the conceptual distinction between robot-independent constant-curvature kinematics and robot-specific actuation mappings [1]. Equations labeled as implementation equations describe this repository and should not be attributed wholesale to that review.

The repository's base Git revision at documentation time was

```
e3e976c97e63596887ccb22e03ad33523c25db72
```

The working tree contained additional modified and untracked files, including the latest topology correction. Consequently, that revision alone does not reproduce this snapshot. A durable research release should commit the intended source, USD assets, documentation, and validation records together and record the resulting revision and asset hashes.

The numerical excerpts in Appendix B are copied from available test output. They are not averages across repeated trials. Historical changes without a retained quantitative test are described as development events, not established performance improvements.

3 Development history and disposition

Development stage	Purpose and current disposition
Initial limb control	Investigated initial bending, loss of commanded pose, and returning toward rest. Holding position, stiffness, and damping became recurring requirements. Early intermediate values are not reconstructed here without retained evidence.
Saved limb and quadruped	Requested new USD versions and reuse in a soft quadruped. A ROS publisher failure exposed use of a dictionary for `_publishers`, which conflicts with rclpy's internal publisher list. Ground hanging constraints, body curvature, and jelly-like motion were also addressed during earlier development. The quadruped remains a separate, historical implementation, not a validated deployment of the current limb.
Hanging limb and launch	Moved the fixed base into the air and hung the limb downward. Corrected the workflow around missing installed loader paths by rebuilding and locating scripts through the installed package.
Control alternatives	Separated position control from hybrid control: phi position versus phi torque, with theta selecting the bending plane. Several direction-range and correction attempts were rolled back. Hybrid control remains preliminary.
Constant curvature and frames	Used the continuum-kinematics review to guide bending conventions. Disc-axis visualization was requested, added, and subsequently rolled back. Disc axes are not an active feature. The intersection of disc radial lines was raised as a geometric accuracy issue.
Unexpected startup bend	Investigated possible lingering ROS controllers after forced VS Code termination. Startup targets are zero, but live external commands can still bend the limb; zero startup targets are not command isolation.
Working-file policy	Established a rule to make subsequent implementation changes in files named `working`, preserving older files as backups.
Virtual sensing	Added six pose-derived virtual gauges that respond to actual motion, including manual manipulation without a controller.
Follower and RViz	Initially used a second simulated limb; retained that version as an in-simulation backup, then replaced the active follower with sensor-driven RViz reconstruction. Added robot description, joint states, robot-state publisher, and a map-to-root transform.
Visualization and plots	Addressed RViz/Snap library conflicts and the missing map frame. Added six live strain plots and portable ZIP/build instructions in the README.
Holding and accuracy	Tuned acceleration drives, body damping, solver settings, and gravity compensation to improve holding in the working limb. This does not establish physical material stiffness.
Six-disc transition	Reduced twelve discs to six, after rollback/reapplication, and generated dedicated six-disc assets. An asset-generation crash motivated native Isaac generation rather than relying on a renderer-free flattened asset.
Shorter spacing	Reduced a temporary 55 mm pitch to 25 mm, changing clear spacing from 50 mm to 20 mm while retaining six discs. Updated geometry, anchors, gains, and RViz spacing together.

Development stage	Purpose and current disposition
Topology correction	The user measured 81.8181568° for a requested 180°. Identified the eleven-versus-five normalization mismatch; added graph-identified legacy position conversion and verified the sixth disc.

This chronology is thematic rather than a timestamped commit log. Rolled-back experiments must not be presented as features of the current configuration.

4 Simulation model and current geometry

The limb is a finite rigid-disc/virtual-link chain with compliant joint drives. It is not a finite-element model of a continuous elastomer and does not provide a resolved material stress or strain field. Continuum terminology describes its intended shape approximation and sensing model.

Quantity	Symbol	Working value
Disc count	N	6
Disc-to-disc interval count	$M = N - 1$	5
Disc thickness	t	0.005 m
Clear spacing	g	0.020 m
Disc-center pitch	$d = t + g$	0.025 m
Straight base-to-tip nominal length	$L_0 = Md$	0.125 m
Gauge/disc radius	r	0.020 m
Hanging generator base elevation	z_0	0.400 m
Runtime disc mass constant	m_d	0.010 kg
Virtual-link mass constant	m_v	0.010 kg
Virtual-link separation	h	0.0005 m

Table 2: Source configuration, not measured hardware dimensions or calibrated material properties. Mass constants are also used in gravity compensation; asset mass agreement should be checked for each research release.

The hanging generator places disc i at $\mathbf{p}_i = (0, 0, z_0 - id)$ in the straight configuration. The base is kinematic; the subsequent discs are movable. Numbering is zero-based: the first disc is `Disk_00`, and the sixth/final disc is `Disk_05`, not `Disk_06`. The Isaac final disc is red; the current RViz final disc is gray. Color does not identify an independent measurement channel.

The first interval has one base D6 joint. Each of the other four intervals has two serial D6 joints, A and B, connected through a virtual link. There are therefore nine active bending joints, not five individual D6 joints. Translation and local rotZ are locked; rotX and rotY are driven. The working topology definition and stage guard require exactly the six intended discs. The rigid-disc view uses explicitly ordered paths, and tip sensing explicitly selects index 5.

5 Constant-curvature reference kinematics

5.1 Ideal continuous section

For a planar constant-curvature section of arc length L, bending magnitude Φ, and bending-plane direction Θ,

$$\kappa = \frac{\Phi}{L}, \qquad R = \frac{1}{\kappa}, \qquad \mathbf{a}(\Theta) = (-\sin\Theta, \cos\Theta, 0)^T. \tag{1}$$

Angles in equations are radians. A reference convention with straight extension along positive local Z has

$$\mathbf{p}(s) = \begin{bmatrix} (1-\cos(\kappa s))\cos\Theta/\kappa \\ (1-\cos(\kappa s))\sin\Theta/\kappa \\ \sin(\kappa s)/\kappa \end{bmatrix}, \tag{2}$$

$$\mathbf{R}(s) = R_z(\Theta)R_y(\kappa s)R_z(-\Theta). \tag{3}$$

The $\kappa \to 0$ limit is a straight section. The hanging implementation uses negative-Z translations, so position signs must be mapped consistently when comparing it with this positive-Z reference convention.

Theta denotes the plane of bending, not an independently commanded physical twist. Total phi is accumulated tangent-orientation change along the disc chain, not the inclination of the base-to-tip chord. For an ideal planar 90° arc, that chord is inclined approximately 45° from the initial tangent.

5.2 Radial-line intersection criterion

For a circular planar centerline, each disc plane should be normal to the local tangent. The radial line *in the bending plane* through each disc center then passes through one common center of curvature. Arbitrary lines within the disc plane do not satisfy this criterion. A straight limb has no finite center of curvature, and nonuniform curvature generally has no single center.

A future quantitative check should fit a circle to measured disc centers, test disc-normal/tangent agreement, and report distances from that fitted center to the corresponding radial lines. Visual agreement alone is not an adequate numerical constant-curvature test.

5.3 Discrete model versus an inextensible continuous arc

The RViz URDF implements each interval through three revolute coordinates, with the composition

$$T_i = R_z(\theta_i)T_z(-d/2)R_y(\beta_i)T_z(-d/2)R_z(-\theta_i). \tag{4}$$

Its transmitted coordinates are $(\theta_i, \beta_i, -\theta_i)$. This symmetric construction provides a disc orientation and hinge-based position, but must not be confused with an exact continuous arc of fixed length d. Its disc-center chord length is

$$c_i = d\cos(\beta_i/2). \tag{5}$$

For a uniform planar bend, its equivalent circular radius is

$$R_{\text{hinge}} = \frac{d}{2}\cot(\beta_i/2), \qquad \ell_{\text{equivalent}} = \beta_i R_{\text{hinge}}, \tag{6}$$

whereas an inextensible constant-curvature interval would have

$$R_{\text{arc}} = d/\beta_i, \qquad c_{\text{arc}} = 2(d/\beta_i)\sin(\beta_i/2). \tag{7}$$

These agree only in the small-angle limit. Thus, fixed nominal pitch and equal orientation increments alone do not prove fixed arc length during bending. The virtual-link offsets and physics leader geometry introduce further details requiring separate positional validation.

6 Position control and holding behavior

6.1 Distribution over the active topology

For total commanded bend Φ_c, the working controller calculates

$$\beta = \frac{\Phi_c}{M} = \frac{\Phi_c}{5}, \qquad \beta_{\text{base}} = \beta, \qquad \beta_{A,j} = \beta_{B,j} = \frac{\beta}{2} \quad (j = 1, \ldots, 4). \tag{8}$$

For 180°, this is 36° at the base and 18° at each split joint. The algebraic sum of all nine scalar bend targets is 180°. These A/B values remain the ROS compatibility representation. Before PhysX initialization, the loader deactivates the eight authored serial A/B constraints and creates one direct D6 joint between each adjacent disc pair. Both A and B half-angle callbacks address that same direct joint; their vectors are summed, so the physical runtime target is β at every one of the five intervals. The unused virtual links are retained as kinematic compatibility objects. The two half-angle vectors are summed before writing the shared direct drive. This removes the serial-pair high-curvature constraint while preserving the existing controller topics. The position controller clamps total requests to ±180°. Runtime X/Y D6 limits are ±80° so the torque-admittance mode can exceed 180° without clipping the base interval.

Each drive receives the implementation mapping

$$q_x = -\beta_j \sin\Theta, \qquad q_y = \beta_j \cos\Theta. \tag{9}$$

This is not generally equivalent to one exact axis-angle rotation at finite angles. Physical joint coordinate coupling matters for diagonal bends; Section 12 reports the observed limitation.

6.2 Drive and solver configuration

Parameter	Working runtime value
Drive type	Acceleration (mass-normalized)
All five direct-joint stiffness / damping	6400 / 240
Linear / angular body damping	2.0 / 10.0
Physics frequency	240 Hz
Position / velocity solver iterations	32 / 8
Controller publishing frequency	20 Hz
Compatibility half-target slew limit	15°/s
Effective direct-interval slew limit	30°/s
Torque magnitude limit	1.5 N m (software bound)
Torque slew limit	2.0 N m/s (software bound)

These are simulator drive settings, not measured Young's modulus, beam rigidity, or a calibrated physical damper. The torque bounds are not hardware-safe operating specifications. The earlier 10^6 N m software bound and 1000 N m example were removed because they drove the model directly to its angular stops and concealed changes in commanded torque.

In position mode, the loader applies upward forces equal to the configured disc and virtual-link weights, using $g = 9.81$ m/s^2. This explicitly compensates gravity; the reported holding test

should therefore not be described as uncompensated, payload-bearing behavior. Compensation is enabled in both position and calibrated torque-admittance modes. Consequently, neither mode is an uncompensated hanging-load experiment.

Targets are ramped locally rather than corrected instantaneously. Since base and split targets have different magnitudes but share the same local slew rate, uniform final targets do not guarantee uniform curvature during the transition. No settling-time sweep or transient constant-curvature proof has yet been recorded.

7 Confirmed twelve-to-six-disc normalization failure

The preserved original position controller has eleven intervals. If it publishes a 180° total request to a five-interval leader, the final disc receives only

$$\Phi_{\text{actual}} \simeq 5\left(\frac{180^\circ}{11}\right) = 81.81818^\circ. \tag{10}$$

The user's reported tip reading was 81.81815680578234°, numerically consistent with this mismatch. The result is a command-normalization problem, not evidence that additional stiffness is required.

The working loader now identifies the known position controller through the ROS publisher graph. A complete contiguous set of A/B publisher pairs is used to distinguish five-interval from eleven-interval sources. For an identified legacy source,

$$q_{\text{adapted}} = q_{\text{received}}\frac{11}{5}. \tag{11}$$

Scaling happens before the local position clamp. A warning reports conversion. The working five-interval controller is not scaled again, and torque targets are not adapted. Manual publishers with a different identity are unchanged.

The local source-built Humble rclpy delivers subscription messages without publisher-message metadata. The adapter therefore requires an unambiguous publisher for each topic and verifies node identity through graph endpoints. Incomplete discovery is retried. Multiple publishers or duplicate controller identities make conversion ambiguous; a warning/error is issued and automatic conversion is disabled. This is compatibility handling, not a guarantee against all conflicting ROS commands. Run one controller only. It recognizes the specified original topology and controller identity, not arbitrary remapped or renamed controllers.

8 Torque-commanded admittance control: implemented interface

The hybrid script publishes a scalar bending-torque command τ_c and a bending-plane direction Θ. The current loader implements a calibrated quasi-static admittance law,

$$\Phi_{\text{eq}} = 180^\circ\frac{\tau_c}{1.0\ \text{N m}}, \qquad |\tau_c| \le 1.0\ \text{N m}. \tag{12}$$

The resulting equilibrium angle is distributed exactly as in position mode: one fifth at each direct disc-to-disc runtime joint. The ROS compatibility arrays still carry one tenth on both A and B topics, and the loader vector-sums the two halves into the full direct-interval target. The same strong acceleration drives then enforce the requested uniform-curvature equilibrium. Torque command slew is limited to 2.0 N m/s.

The command scale was increased by a factor of 100 from the earlier 0.010 N m-per-180° mapping. The current 0–1.0 N m range is approximately 0–10.2 kgf cm and is therefore easier to compare with compact actuator ratings. This rescaling changes only the admittance interface and not the simulated equilibrium motion; it is not an experimentally identified moment–curvature law or a claim that a physical limb requires this torque.

This is a *torque-commanded admittance controller*, not raw open-loop body-torque actuation and not a measured physical constitutive law. Earlier raw-torque diagnostics produced 3.5555° at 0.001 N m and 7.1150° at 0.002 N m, but curvature was nonuniform and depended on gravity and effective joint inertia. A temporary low-stiffness force-spring trial also concentrated curvature toward distal intervals. Those approaches were rejected because they did not reproduce the required constant-curvature shape.

The active rescaled direct-topology bounded tests measured 18.0000°, 90.0000°, and 179.9999° at 0.1, 0.5, and 1.0 N m. All five interval rotations were respectively approximately 3.6°, 18°, and 36°. Negative torque produces the opposite signed target. These values verify the implemented software mapping; they must not be interpreted as hardware torque characterization. An explicit over-limit regression requested 1.5 N m; the controller applied 1.0 N m and measured 179.9999°, with 0.0001° mapping error and approximately 36° at every interval. This confirms the active 180° cap in the tested path.

The prior serial A/B topology reached only approximately 262.3° for a 270° request and stopped near 52° per later disc interval. Collision filtering, increased drive gains, and asymmetric A/B targets did not remove this ceiling and were not retained. Replacing each runtime serial pair with one direct disc-to-disc D6 joint produced a historical 270.0000° result, with all five intervals at approximately 54.0000°. That development result is retained as historical evidence; the active command is clamped to 1.0 N m and 180°.

While a nonzero torque command is active, incoming bend-position callbacks are ignored, preventing a stale 20 Hz position publisher from zeroing the torque command. The loader logs a warning when it first rejects a position command. Publishing zero torque releases this priority. One controller at a time remains the required experimental practice.

9 Six virtual strain gauges

9.1 Placement and retained topic names

Station	Disc	s/L_0	Angles about circumference
`one_third`	`Disk_02`	0.40	0°,120°,240°
`two_thirds`	`Disk_03`	0.60	180°,300°,60°

The original placement requirement was one-third and two-thirds of the limb length. With six discs, exact thirds lie between discs; the working rings use the nearest symmetric pair at 40% and 60%. Topic names are retained for compatibility. The second set is offset by 180°; threefold 120° symmetry makes the combined rings appear staggered by 60°.

9.2 Measured orientation to local curvature

Let q_i be the normalized disc orientation, represented as a quaternion in $[w, x, y, z]$ order. At sensor disc i, the implementation forms

$$q_{\mathrm{rel},i} = q_{i-1}^{-1} q_{i+1}, \qquad \boldsymbol{\omega}_i = \mathrm{rotvec}(q_{\mathrm{rel},i}). \tag{13}$$

Quaternion sign is chosen so that the scalar part is nonnegative; the rotation vector consequently uses a shortest-path rotation. The local curvature estimate is

$$\hat{\kappa}_{x,i} = \frac{\omega_{x,i}}{2d}, \qquad \hat{\kappa}_{y,i} = \frac{\omega_{y,i}}{2d}, \qquad 2d = 0.050 \text{ m}. \tag{14}$$

For the first ring this uses discs 1 and 3; for the second it uses discs 2 and 4. Z rotation is not used in the bending-strain estimate. These are nominal-spacing finite-difference estimates, not differentiation of a measured deformable centerline. The orientation difference is expressed relative to the preceding disc; interpretation as centered curvature is approximate for general nonplanar bending.

9.3 Curvature to virtual axial strain

For a gauge angle α, its nominal coordinates are $x = r\cos\alpha$, $y = r\sin\alpha$. The implemented strain convention is

$$\hat{\varepsilon}_i(\alpha) = -\hat{\kappa}_{x,i} y + \hat{\kappa}_{y,i} x = r\left(\hat{\kappa}_{y,i}\cos\alpha - \hat{\kappa}_{x,i}\sin\alpha\right). \tag{15}$$

This is dimensionless. The published reading is

$$\hat{\varepsilon}_{\mu\varepsilon} = 10^6 \hat{\varepsilon}. \tag{16}$$

For example, 10,000 $\mu\varepsilon$ corresponds to a strain magnitude of 0.01, or 1%. Positive and negative readings represent opposite bending sides under the implementation's coordinate convention; their correspondence to physical tensile/compressive polarity must be checked against a hardware gauge installation and longitudinal-axis convention.

For ideal uniform curvature, substituting $(\kappa_x, \kappa_y) = \kappa(-\sin\Theta, \cos\Theta)$ gives

$$\varepsilon(\alpha) = r\kappa\cos(\alpha - \Theta). \tag{17}$$

The ideal three-gauge ring has zero summed bending strain. A theoretical 90° bend over nominal length 0.125 m yields $r\kappa \approx 0.2513$, or about 251,327 $\mu\varepsilon$ at the extreme gauge position. This large value is a consequence of the assumed radius and kinematic equation, not a calibrated prediction for a practical gauge or proof that a real material can sustain that deformation. It is a calculated illustration, not a retained simulation measurement.

9.4 Independence from command and physical limitations

Readings use live disc poses, not desired phi or theta. They therefore respond to actual simulated motion and manually changed orientations without requiring a controller. The gauge markers are visual objects; they do not measure a deforming material surface.

The model excludes measured axial extension, shear strain, torsional strain, constitutive behavior, finite-strain calibration, noise, thermal drift, hysteresis, gauge bonding, and sensor latency. Equation (15) is an idealized kinematic emulation, not an experimentally validated strain field. A material law is not inferred from the drive stiffness. Use the term *orientation-derived virtual bending strain* in research reporting.

9.5 Virtual Wheatstone-bridge layer

The current working implementation additionally converts each of the six virtual strain values to an ideal independent quarter-bridge voltage. Its explicit research assumptions are nominal resistance

$R_0 = 350\ \Omega$, gauge factor $GF = 2.0$, and excitation $V_{\text{exc}} = 2.5$ V. If the active gauge occupies the selected bridge arm, its modeled resistance is

$$R_g = R_0(1 + GF\varepsilon), \tag{18}$$

and its differential output is

$$V_{\text{out}} = -V_{\text{exc}} \frac{GF\varepsilon}{4 + 2GF\varepsilon}. \tag{19}$$

The sign is a bridge-wiring convention and must be matched to the eventual physical connection. Near zero strain, $V_{\text{out}}/V_{\text{exc}} \approx -GF\varepsilon/4$; thus 1000 $\mu\varepsilon$ produces approximately -0.5 mV/V, or -1.25 mV at 2.5 V excitation.

The voltage is published even when it is only a circuit extrapolation, while separate validity flags are false when $|\varepsilon| > 3\%$ or $|V_{\text{out}}/V_{\text{exc}}| > 25$ mV/V. A value with false validity must not be reported as a feasible physical foil-gauge measurement. At compressive strain $\varepsilon \leq -1/GF$, modeled active resistance is nonpositive and the voltage is published as NaN. The working model's predicted high-bend strains often exceed the 3% flag. Consequently, this layer provides a traceable voltage-interface hypothesis, not a validated selection of a real gauge or a demonstration that a foil gauge can survive direct mounting.

Matplotlib normally breaks a trace at a NaN sample. For visualization only, the working voltage plot maps these non-finite samples to a dashed $+V_{\text{exc}}/2 = +1.25$ V invalid-model boundary and emits a warning. The published ROS value remains NaN and the validity flag remains false, avoiding the false claim that the boundary is a feasible measurement. The voltage plot uses 0.02 V minimum vertical padding, rather than the strain plot's 1.0-unit padding, so finite changes from 1.0 to 1.25 V remain discernible.

10 Sensor-driven reconstruction and RViz

Each ring supplies three strain values. Inverting Equation (15) uses the matrix

$$\mathbf{A} = r \begin{bmatrix} -\sin\alpha_1 & \cos\alpha_1 \\ -\sin\alpha_2 & \cos\alpha_2 \\ -\sin\alpha_3 & \cos\alpha_3 \end{bmatrix}, \qquad \hat{\boldsymbol{\kappa}} = \arg\min_{\boldsymbol{\kappa}} \|\mathbf{A}\boldsymbol{\kappa} - \boldsymbol{\varepsilon}\|_2^2. \tag{20}$$

Published microstrain is converted back to dimensionless strain first. The three readings estimate two bending components; a single ring is not a six-degree-of-freedom pose sensor. Axial stretch or torsion is not estimated.

The two recovered curvature vectors are interpolated componentwise at interval midpoints. NumPy interpolation uses the nearest endpoint value outside the two ring locations. Each interval is assigned

$$\beta_i = d\|\hat{\boldsymbol{\kappa}}_i\|, \qquad \theta_i = \operatorname{atan2}(-\hat{\kappa}_{x,i}, \hat{\kappa}_{y,i}). \tag{21}$$

The generated six-disc URDF contains fifteen movable visualization joints: three per interval as in Equation (4). The reconstructed joint states are published to `/joint_states`; robot-state publisher converts them to TF using the supplied robot description. A static transform connects `map` to `soft_limb_disk_00`.

The RViz limb receives sensor-derived joint states, not the leader controller's commands. Nevertheless, since its inputs are generated from leader poses through the same strain equation used for inversion, reconstruction agreement is partly an internal consistency test. It is not independent validation of the strain model. Only two rings are available; localized bends outside or between those rings cannot generally be recovered uniquely.

11 ROS interface and plotting

Stream	Purpose and units
Joint `roll/position`	Local bend target in degrees; base and pairs 01–04 A/B. Identified legacy targets are converted by the adapter.
Joint `yaw/position`	Bending-plane direction in degrees, wrapped around the circumference.
Joint `roll/torque`	Scalar torque command in N m, interpreted through the calibrated admittance law.
`measured_tip/phi`	Norm of the sum of adjacent-disc local X/Y rotation vectors, converted to degrees.
`measured_tip/theta`	Direction from that accumulated vector in degrees.
`measured_tip/orientation`	Relative tip quaternion $[w, x, y, z]$.
Ring `strain`	Three-element `Float64MultiArray`, microstrain.
Ring `gauge_1..3/strain`	Individual scalar microstrain channels.
Ring `bridge_voltage`	Three-element ideal quarter-bridge output in V.
Ring `bridge_valid`	Three-element 1.0/0.0 nominal-range flag.
Ring `gauge_1..3/bridge_voltage`	Individual scalar voltage channels.
Ring `gauge_1..3/bridge_valid`	Individual Boolean validity channels.
Ring `orientation, phi, theta`	Ring pose relative to the base; quaternion and derived angles.
`/robot_description`	Generated URDF for robot-state publisher/RViz.
`/joint_states`, `/tf`	Reconstructed kinematic joint states and visualization transforms.

The corresponding topic prefixes are:

- joint commands: `/soft_limb/pair_XX/...`;
- sensor rings: `/soft_limb/sensors/<station>/...`; and
- measured tip: `/soft_limb/measured_tip/...`.

The working launch starts two Matplotlib plotters: one subscribes to all six individual strain channels and one to all six bridge-voltage channels. Each uses two rows of three plots, a rolling 20 s history, and a 20 Hz refresh limit. The horizontal axis uses local monotonic wall time, not a synchronized simulation timestamp. Plot histories are not automatically exported as a research dataset. Timestamp alignment, recording, and replay need to be added or managed through a documented logging workflow.

The station orientation topics publish each instrumented disc's quaternion relative to the base. During the September 24, 2026 full-launch regression, Isaac Sim initialized but the loader then exited because this relative quaternion was referenced before assignment in the sensor-publication path. The assignment was restored, and the headless verification harness was extended to execute the complete pose, strain, bridge-voltage, and station orientation publication path so this class of launch-only error is detected. The corrected installed loader then remained active for a 35 s headless full-entry-point test. That test also exposed a shutdown-only race: a signal could invalidate

the ROS context during rendering before the next sensor publication. The loop now checks ROS context validity at entry and immediately before publishing, preventing this misleading termination traceback. A repeated 20 s test of the rebuilt installed loader reached the simulation loop and ended at the intentional timeout without a Python traceback. The expanded sensor-publication verifier also passed at 0.005 N m, with an expected bend of 90.0000°, measured final-second range of 90.0000–90.0000°, and reported tracking error of 0.0000°.

12 Recorded preliminary verification

The direct runtime topology requires special treatment during command interpolation. The base interval receives one command, whereas each of the other four physical intervals receives the vector sum of compatibility commands A and B. Slewing both halves at the full 20°/s produced a 40°/s physical target away from the base and only 20°/s at the base. Although the final equilibrium remained uniform, this caused a visibly segment-wise transient. Each compatibility half is now limited to 10°/s, so all five physical interval targets initially advanced together at 20°/s. The synchronized rate was subsequently increased to 30°/s by using 15°/s for each compatibility half. This preserves the commanded constant-curvature distribution during the bend while reducing transition time. The verification harness explicitly compares all five X/Y drive targets after the first update. An eight-second 180° Isaac regression at the 30°/s setting reported identical first-step targets $[0.0, 0.125]$ degrees on all five direct joints, a final measured range of 179.9999–179.9999°, and approximately 36° at every interval (PASS).

12.1 Method and metrics

The headless harness opens the hanging six-disc asset, validates topology, applies the working drive/solver configuration, and receives commands through the actual ROS position controller. Its default run is 12 s of simulated time at 240 Hz (2880 physics steps); it measures the last simulated second. It does not start the complete RViz/plotter launch. Thus it is an end-to-end command-to-physics test, not a full graphical-launch validation. It now exercises the complete sensor-publication method, although it still does not validate the RViz and Matplotlib graphical processes themselves.

For position tests through 180°, the target tip quaternion is constructed from axis $\mathbf{a}(\Theta)$ and angle Φ_c. Full orientation error is

$$e_R = \left\| \mathrm{rotvec}(q_{\mathrm{target}}^{-1} q_{\mathrm{tip}}) \right\|, \tag{22}$$

reported in degrees. Reported phi instead sums the local X/Y rotation vector $\mathrm{rotvec}(q_i^{-1} q_{i+1})$ over all five disc intervals and takes the norm of the result. Each local increment remains on its shortest quaternion path, but their sum can represent an accumulated bend above 180°. The raw published tip quaternion remains orientation-only and cannot by itself distinguish, for example, 0° from 360°.

Harness tolerances are 2° full orientation error and a 1° peak-to-peak phi range during the final second. These are software-test tolerances, not calibrated measurement uncertainty. Isaac fast shutdown can obscure a raised exception in the process exit status; inspect the explicit `VERIFY RESULT: PASS/FAIL` and retain the output rather than relying on exit status alone.

The successful tests printed a final-second range of 179.9996–179.9996° at four decimal places. This establishes no visible variation at the reported precision under those conditions, not mathematically zero oscillation. Adjacent-disc orientation increments were approximately 36° in the single-axis test. Matching orientation increments does not by itself validate disc-center geometry, payload robustness, hardware behavior, or all-direction performance.

Case	Final phi	Full tip error	Status
Working controller, 180°, theta=0	180.0001°	0.0001°	PASS
Legacy controller with adapter, same target	179.9996°	0.0004°	PASS
Torque admittance, 0.1 N m	18.0000°	0.0000° mapping error	PASS
Torque admittance, 0.5 N m	90.0000°	0.0000° mapping error	PASS
Torque admittance, 1.0 N m	179.9999°	0.0001° mapping error	PASS
Active cap, 1.5 N m requested	179.9999°	Applied 1.0 N m	PASS
Historical pre-cap torque trial, 1.5 N m	270.0000°	0.0000° mapping error	PASS
Large diagonal, 180°, theta=45°	About 178.93°	About 10.25°	Limitation
User observation before compatibility fix	81.8181568°	Not recorded	Mismatch

Table 4: Individual recorded results. The diagonal result is retained from the prior working drive-map test and README, not a new diagonal trial of the latest compatibility adapter. No repeated-trial statistics are claimed.

Six offline adapter regressions passed: legacy scaling, working no-double-scale, manual publisher preservation, incomplete graph detection, duplicate controller ambiguity, and the local Humble message-only callback. A real ROS test using the original publisher also confirmed base 36° and an algebraic total of 180° across nine active drives. The ROS package rebuilt successfully after the correction.

13 Third objective: three-module hanging snake

The verified single-module design was extended into a serial hanging robot named `soft_3_working`. The configuration contains three six-disc modules (18 physical discs total), five actuated intervals per module (15 actuated intervals total), and two fixed inter-module connectors. The uppermost disc is kinematic, leaving the complete chain suspended under gravity. Each module receives its own bending-torque command $\tau_{c,j}$ and bending-plane direction θ_j, for $j \in \{1, 2, 3\}$. The generator, loader, controller, plotter, verifier, launch, and USD filenames use the same shortened `soft_3_working` naming convention. The established `/soft_snake/module_N` ROS topic namespace is retained to avoid breaking recorded topic-based experiments. The existing active admittance law is applied independently,

$$\phi_j = 180^\circ \frac{\tau_{c,j}}{1.0\ \mathrm{N\,m}}, \qquad |\tau_{c,j}| \le 1.0\ \mathrm{N\,m}, \tag{23}$$

with $|\phi_j| \le 180^\circ$. Within each module, the total target is distributed uniformly over its five physical intervals and all five targets use the same slew rate. The connectors are not assigned a bend command, so they transmit the terminal pose of one module to the base of the next.

The sensing layout is repeated locally on every module. One ring of three gauges lies at one-third module length at 0°, 120°, and 240°; a second ring lies at two-thirds length at 180°, 300°, and 60°. Thus the serial robot exposes 18 strain channels, 18 ideal quarter-bridge voltage channels, and 18

validity flags. The per-gauge namespace is `/soft_snake/module_N/sensors/<station>/gauge_G/<quantity>`. The presentation records the 18 bridge-voltage channels rather than repeating the strain plot. Simulator screenshots use Isaac Sim's *Grey Studio* viewport light rig, which the working loader selects programmatically before the capture. The `soft_3_working` ROS launch now starts the live 18-channel bridge-voltage plotter together with Isaac Sim; plotting begins as soon as sensor messages are available.

A 12-second headless regression commanded $(\tau_{c,1}, \tau_{c,2}, \tau_{c,3}) = (0.50, 0.75, 1.00)$ N m and $(\theta_1, \theta_2, \theta_3) = (0, 90, -90)^\circ$. Measured module bends were 90.0047476°, 134.9999084°, and 179.9998960°, respectively. Corresponding absolute tracking errors were 0.0047476°, 0.0000916°, and 0.0001040°. The final-second ranges were 0.0001642°, 0.0000568°, and 0.0000713°. Residual bends across the two fixed connectors were 0.0000672° and 0.0000394°. All five first-step targets were synchronized within each module, 18 gauge publishers were detected, and the test returned `PASS`. These results validate the software realization, not physical torque or strain calibration.

14 Known limitations and rejected approaches

1. Equation (9) produces a substantial full-orientation error at the recorded large diagonal bend. Exact all-direction constant-curvature tracking is not established, despite successful single-axis tracking.

2. An attempted runtime rotation of both D6 reference frames invalidated physics handles in the test and was removed. That strategy is not active.

3. Several earlier corrections and disc-count changes were explicitly rolled back. Their intended benefits must not be treated as retained results.

4. Equal final orientation increments do not prove continuous fixed-length curvature, and local slew limits may produce nonuniform transition shapes.

5. Gravity-compensated holding does not establish loaded or unassisted holding performance. Contact, disturbance, and mass uncertainty need testing.

6. Strain estimation and reconstruction share an idealized equation; neither has been validated against real gauges or an independent material model.

7. ROS graph-based conversion depends on publisher identity and discovery. Stale/duplicate controllers and nonstandard remappings remain operational risks.

8. Two sensor rings cannot uniquely recover arbitrary distributed curvature, torsion, axial stretch, or full shape without additional assumptions.

9. The current direct-topology torque-admittance law has been verified only for unloaded, single-axis simulation. The active rescaled interface spans 0.1 to 1.0 N m through a maximum of 180°. It is not raw torque control, a material constitutive model, or a hardware calibration; negative commands, payloads, disturbances, and arbitrary bending directions require separate validation. The quadruped and two-physical-limb backup were not part of these tests.

15 Reproducible operation and asset handling

15.1 Environment

The working environment uses Ubuntu 22.04, ROS 2 Humble, Isaac Sim 4.5.0, an NVIDIA GPU/-driver, colcon, RViz2, robot-state publisher, TF2, and Python Matplotlib/Tk. One successful diagnostic environment reported an RTX 3080 and driver 580.178.04; this is a test environment, not a universal prerequisite. For each recorded experiment retain exact versions, USD hashes, timestep, solver settings, drive values, and whether gravity compensation was enabled.

Paths below are portable examples. Replace the ROS setup path for a source-built installation and set the actual Isaac installation location. Run from the extracted repository root named `soft_world`.

```
cd /path/to/soft_world
source /opt/ros/humble/setup.bash
export ISAAC_SIM_PATH=/path/to/isaacsim
colcon build --packages-select soft_world
source install/setup.bash
ros2 launch soft_world soft_world_working.launch.py
```

In a separate terminal, source the same ROS and workspace setup, then run

```
cd /path/to/soft_world
source /opt/ros/humble/setup.bash
source install/setup.bash
python3 scripts/control_soft_limb_working.py --ros-args \
  -p phi_position:=180.0 -p theta_position:=0.0
ros2 topic echo /soft_limb/measured_tip/phi --once
```

The echo is run in another terminal while the controller is active. Default working position parameters are phi=90° and theta=0°; explicit experimental parameters are preferable.

To verify independently from the graphical launch, stop other controllers, source the environment, and run

```
python3 scripts/test_position_topology_working.py
"$ISAAC_SIM_PATH/python.sh" scripts/verify_bend_working.py \
  --phi 180 --theta 0
"$ISAAC_SIM_PATH/python.sh" scripts/verify_bend_working.py \
  --phi 180 --theta 0 --legacy-controller
```

Retain stdout/stderr and the explicit verification status. Run simulations sequentially to avoid resource contention. These commands require functioning GPU access and permission to write runtime ROS/Isaac logs.

15.2 RViz and safe operation

Use fixed frame `map`, add RobotModel, choose description source Topic, and set description topic `/robot_description`. The working launch publishes the map-to-base transform. Its RViz/plotter processes remove selected Snap-injected GTK/GIO paths to avoid the observed host-library ABI conflict; this workaround is environment-specific.

Start only one controller after the simulation loads. Stop the controller before stopping the launch. A forced VS Code exit may leave controllers alive; inspect nodes and process identity before terminating a confirmed stale process. Restart the simulator after loader changes; a running Python process does not automatically reload an edited installed script.

15.3 USD generation, portability, and version control

The active launch loads `usd/soft_limb_hanging_6_disc_working.usd` directly. It does not prune a twelve-disc scene at runtime. A matching six-disc position asset is retained separately from older twelve-disc assets. Native generators are invoked by `scripts/create_six_disc_assets_working.py`; they require Isaac's initialized schema/runtime environment. The authored geometry updater preserves existing native metadata rather than rebuilding a flattened asset without that environment. Asset references must be inspected when distributing to another machine; portable Python paths do not alone establish portable USD references.

For ZIP sharing, retain source, launch files, package metadata, USD assets, README, and this research documentation; omit generated `build/`, `install/`, `log/`, and caches. Rebuild at the recipient's location. `.gitignore` only governs untracked files in Git; it does not delete files, untrack committed assets, or filter a manually created ZIP. Preserve the working-file policy and avoid replacing historical backups.

16 Recommended next research experiments

These are proposed tests, not completed results.

1. Run repeated position trials across phi, theta, and both bend signs. Report quaternion error, signed interval angles where definable, final phi range, overshoot, settling time, and run-to-run variation.

2. Compare measured disc centers and orientations with a fitted circle and the continuous fixed-length reference. Report radial-line intersection residuals and distinguish hinge geometry from true continuous-arc geometry.

3. Sweep damping, drive gains, solver iterations, timestep, gravity compensation, and applied load separately. Control these variables rather than attributing all angle errors to stiffness.

4. Test localized manual perturbations and known nonuniform curvature. Compare sensor-driven reconstruction against independent pose ground truth, especially outside the sensor rings.

5. Validate virtual strain against a separate deformable model or hardware gauges, including sensor radius, sign, axial extension, finite-strain behavior, gauge factor, noise, calibration, and temperature sensitivity as appropriate.

6. Select the physical limb material and maximum curvature before purchasing foil gauges. A 350 ohm linear foil gauge and a six-channel quarter-bridge DAQ are a candidate architecture, but the simulated 25–50% high-bend surface strain is outside ordinary foil-gauge assumptions.

7. Introduce synchronized simulation timestamps, durable datasets, and experiment manifests. Preserve raw data and analysis code, not only screenshots or rolling plot displays.

8. Validate hybrid effort equilibrium and torque distribution separately; only then consider hardware transfer or quadruped integration.

Potential research questions include whether six-disc reconstruction error remains acceptable relative to a twelve-disc reference, how sparse gauge rings limit shape observability, and whether geometry-aware actuation can reduce the large-angle diagonal error without sacrificing stable compliant motion.

17 Conclusion

The working platform supports a six-disc hanging leader, separate position and calibrated torque-admittance interfaces, six pose-derived strain channels, live graphs, and sensor-driven RViz reconstruction. It now also includes the `soft_3_working` serial hanging robot with three independently directed torque-admittance modules and 18 pose-derived gauges. The confirmed eleven-versus-five normalization failure was corrected in the loader and reproduced in a successful legacy-controller regression. The latest reported 180.0001° position result and 179.9999° active torque-admittance endpoint are specific software verification results; the earlier 270.0000° result is retained only as a pre-cap development record. Physical strain validity, exact inextensible constant-curvature geometry, arbitrary bending-direction accuracy, and hardware applicability remain open research tasks rather than established conclusions.

17.1 Research progress presentation

An editable progress presentation was created to communicate the work completed after the successful single-module position-control milestone. It includes fresh captures of the current Isaac Sim environment, live six-channel strain and bridge-voltage graphs collected while stepping the torque command through 0, 0.5, and 1.0 N m, quantitative constant-curvature and 180°-cap figures, the ROS/RViz data path, the three-module snake architecture and its 18-channel sensing/validation results, limitations, and proposed next experiments. The source presentation, PowerPoint export, PDF export, and captured assets are stored under `research_documentation`; plotted sensor values are explicitly labeled as simulated and pose-derived rather than hardware-validated.

References

[1] R. J. Webster III and B. A. Jones, "Design and kinematic modeling of constant curvature continuum robots: A review," *The International Journal of Robotics Research*, vol. 29, no. 13, pp. 1661–1683, 2010. doi:10.1177/0278364910368147.

[2] SoftWorld working repository: README, working Python implementation, and six-disc USD assets. Inspected September 15, 2026; working snapshot extends the base revision recorded in Section 2.

[3] SoftWorld development conversation and headless/ROS verification output, September 15, 2026. Numerical excerpts retained in this document and `validation_record_working.txt`; not an independently published dataset.

A Active file map

Repository-relative file	Role
`launch/soft_world_working.launch.py`	Simulator, RViz, TF, description, and plots.
`launch/soft_3_working.launch.py`	Three-module hanging snake launch.
`scripts/load_soft_3_working.py`	Three-module physics, control consumption, measurements, and 18-gauge publication.

Repository-relative file	Role
scripts/control_soft_3_working.py	Three independent phi-torque and theta command pairs.
scripts/plot_soft_3_working.py	Eighteen strain or bridge voltage graphs.
scripts/verify_soft_3_working.py	Bounded three-module headless regression.
scripts/soft_3_working.py	Generator for the serial hanging USD.
usd/soft_3_working.usd	Active three-module hanging asset.
scripts/load_soft_limb_working.py	Physics configuration, command consumption, sensing, torque application, and reconstruction.
scripts/control_soft_limb_working.py	Five-interval position command.
scripts/control_soft_limb_hybrid_working.py	Calibrated torque-admittance and direction command.
soft_world/limb_configuration_working.py	Shared six-disc topology, tip index, bend normalization, and stage validation.
scripts/limb_configuration_working.py	Import bridge that avoids the legacy scripts/soft_world.py package-name collision.
scripts/position_topology_adapter_working.py	Identified legacy position conversion.
scripts/test_position_topology_working.py	Six offline adapter regressions.
scripts/verify_bend_working.py	ROS-to-physics headless verification.
scripts/plot_strain_gauges_working.py	Six live plots.
scripts/bridge_model_working.py	Reproducible ideal 350 ohm quarter-bridge voltage and validity model.
scripts/test_bridge_model_working.py	Four offline bridge-model checks.
soft_world/robot_description_working.py	Six-disc, fifteen-joint RViz URDF.
scripts/soft_limb_hang_working.py	Hanging USD generator.
scripts/soft_limb_working.py	Position asset generator.
scripts/create_six_disc_assets_working.py	Native generator wrapper.
scripts/update_six_disc_geometry_working.py	Native authored geometry update.
usd/soft_limb_hanging_6_disc_working.usd	Active hanging asset.
usd/soft_limb_position_control_6_disc_working.usd	Position asset.
scripts/load_soft_limb_in_simulation_working.py	Older two-limb backup.
launch/soft_world_in_simulation_working.launch.py	Backup launch.

B Retained numerical excerpts

Working-controller test:

```
VERIFY controller: base=36.0 deg,
split joints=[18.0, 18.0, 18.0, 18.0] deg;
```

```
tip=/World/SoftLimb/Disk_05
tip_orientation_error=0.0004 deg
last_second_phi_range=179.9996..179.9996 deg
intervals=[35.999966601826934, 35.99982317874544,
35.999893167570946, 35.999921731340976, 35.999989327436786]
VERIFY RESULT: PASS
```

Original twelve-disc controller with the working loader's adapter:

```
Detected legacy 12-disc position controller:
rescaling bend targets by 11/5.
180 degrees now refers to the sixth/final disc Disk_05.
VERIFY controller: base=36.0 deg,
split joints=[18.0, 18.0, 18.0, 18.0] deg;
tip=/World/SoftLimb/Disk_05
tip_orientation_error=0.0004 deg
last_second_phi_range=179.9996..179.9996 deg
VERIFY RESULT: PASS
```

Separate ROS command verification:

```
PASS actual ROS legacy controller:
base=36.000000, total=180.000000
```

The A and B arrays were both checked in the harness; the printed split array above displays A only. Rounded equality of output fields is not a substitute for a time-series dataset or uncertainty analysis.

C Suggested experiment-record template

```
Experiment ID / researcher / date:
Research question and hypothesis:
Git revision and uncommitted changes:
USD asset names, SHA-256 hashes, reference portability:
OS / ROS / Isaac / Python / GPU / driver versions:
Disc count, pitch, radius, masses:
Drive mode, stiffness, damping, body damping:
Timestep, solver iterations, gravity compensation:
Controller identity and parameters:
Loads, contact conditions, manual disturbances:
Simulation duration and timestamp convention:
Raw logs / rosbag / analysis script locations:
Metrics, tolerances, repeated-trial count:
Results, uncertainty, limitations, and disposition:
```

Record any generative-AI assistance in accordance with the researcher's program and publication requirements. The researcher should review derivations, citations, source changes, and claimed evidence before academic submission.